

From supervised learning to our RL training loop

Dr Nguyen's board shows classification as four beats: **predict** → **score** → **differentiate** → **update**. The same beats appear in our trading agent; only the data and the score change.

Supervised objective (his board)

$$\min_{\theta} \frac{1}{N} \sum_{n=1}^N \ell(f(x_n; \theta), y_n)$$

RL objective (Iteration 1 trading)

$$\max_{\theta} \frac{1}{K} \sum_{k=1}^K \sum_{t=0}^{T-1} \gamma^t r_t^{(k)} \quad \text{with} \quad a_t \sim \pi_{\theta}(a_t | s_t), \quad r_t = W_{t+1} - W_t, \quad W_t = C_t + n_t P_t.$$

His board (classification)

```
net = ResNet18()
opt = Adam(...)

# data (x, y) fixed
preds = net(x)          # f(x; theta)
loss = MSE(preds, y)/N

opt.zero_grad()
loss.backward()          # grad loss
opt.step()               # theta <- theta - a*grad
```

Ours (trading / PPO)

```
policy = PolicyNet()      # pi_theta
critic = ValueNet()       # V_phi
opt_pi, opt_V = Adam(), Adam()

# data created by acting
B = ROLLOUT(prices, pi)
A = advantages(B, V)
L_pi = - PPO_CLIP(pi, B, A)
L_V = MSE(V(s), returns)

opt_pi.zero_grad(); L_pi.backward(); opt_pi.step()
opt_V.zero_grad(); L_V.backward(); opt_V.step()
```

Key translation

- Fixed labels y_n become rewards r_t collected online.
- One loss ℓ becomes policy objective L^{CLIP} plus value loss L^V .
- Minimise prediction error becomes maximise expected wealth change (via PPO).
- Dataset independent of θ becomes trajectories that depend on π_{θ} .

Use with Algorithm 3.1. Algorithm 3.1 is the full day-loop. This figure is the compact “how to get the policy” template under **Objective function** → **How to get policy**.